Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/15/2005
PI Name:	Allen, Mark Ph.D.		
Project Title:	Photonic Technologies for Early Detection of Human Disease		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01810	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/08/2002	**End Date:**	05/31/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Allen, Mark		
COI Name (Institution):	Tittel, Frank (Rice University) O'Keefe, Tony (Los Gatos Research) Kosterev, Anatoliy (Rice University) Bakhirkin, Yuri (Rice University)		
Grant/Contract No.:	NAS2-02038		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To develop and demonstrate a biosensor platform technology for non-invasive disease detection and health monitoring based on high-sensitivity detection of trace gases in expired breath. The platform technology combines room-temperature mid-IR lasers(Quantum Cascade Laser - QCL) and resonant optical cavities (Cavity Enhanced Spectroscopy - CES) to realize a portable, consummable-free system for routine astronaut health monitoring Year One (complete): develop complete list of known trace expired breath gases associated with specific pathologies; demonstrate detection of large molecular weight (broadband absorbers)species; compare various CES approaches for optimum detection of small molecular weight species; develop optical design for compact resonant cell. Year Two (complete): demonstrate CES detection of large molecular weight species; demonstrate time-resolved, CES detection of small molecular weight species; develop optical design for multi-wavelength resonant cells. Year Three (in progress): demonstrate time-resolved CES detection of large and small molecular weight gases in expired human breath; demonstrate multi-species monitoring using multi-wavelength cells; develop prototype portable sensor design The project is the first systematic exploration of the various cavity enhanced spectroscopies using the recently available Quantum Cascade laser technology. It compares time-domain to integrated throughput measurements to determin the best method for utilizing the quasi-cw, relatively broadband output of room-temperature QCL's. The final goal will provide for the first time a practical sensor that will allow physicians to have routine access to trace, often reactive, gases in expired human breath.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We extended our results with cw off-axis ICOS of NO to CO using a cryogenically-mounted QCL near 4.6 microns. While these results were encouraging, they could not be directly applied to breath analysis since the laser wavelength was unsuitable. We ordered a second cryogenic laser at the optimum wavelengths identified in previous reports and plans to use this laser with the existing 5.2 micron laser for simultaneous cw-ICOS measurements of both NO and CO in exhaled breath. While waiting for the new 4.6 micron laser, we completed design and assembly of the dual laser setup and evaluated a cw room-temperature laser near 5.2 micron. The room-temperature laser evaluation is important as we expect that room-temperature lasers will eventually supercede the cryogenic devices. We also completed measurements of CO in human breath from 10 static, individual breath samples. For each sample, we measured the cavity-enhanced absorption spectra and fitting them to determine their CO content. All of the breath samples taken were between 800 to 2200 ppbv CO with an average value of 1675 ± 438 ppmv. Note that no smokers were tested, nor were the tests properly standardized to eliminate dietary efforts (e.g., fasting) or assure entitled breath testing (e.g., no exhalation standards). However, the prototype demonstrated its ability to determine CO mixing ratios in actual breath samples without any degradation in performance due to particulates or moisture.
Bibliography Type:	Description: (Last Updated: 09/02/2009)
Articles in Peer-reviewed Journals	Bakhirkin YA, Kosterev AA, Roller C, Curl RF, Tittel FK. "Mid-infrared quantum cascade laser based off-axis integrated cavity output spectroscopy for biogenic nitric oxide detection." Applied Optics, 2004 Apr 10;43(11):2257-66. PubMed PMID: 15098827 , Apr-2004
Presentation	Allen, M.G. "Applications of quantum cascade and interband cascade lasers to chemical sensing from 3 - 100 microns" OSA Topical Meeting on Laser Applications to Chemical and Environmental Analysis Feb-2004
Presentation	McCurdy, M., et al. "Off-axis integrated cavity output spectroscopy for nitric oxide detection in human breath using a quantum cascade laser" American Academy of Allergy, Asthma, and Immunology Meeting Mar-2004
Presentation	Bakhirkin, Y.A., et al. "Wavelength modulation off-axis integrated cavity output spectroscopy for biogenic NO detecion in human breath" OSA CLEO/IQEC 2004 May-2004
Presentation	Wainner, R., et al. "Hydrocarbon gas sensing using 3.5 micron type II interband cascade lasers" OSA CLEO/IQELS 2004 May-2004
Presentation	Silva, M., et al. "Cavity-enhanced spectroscopy using room-temperature quantum cascade lasers" OSA CLEO/IQELS 2004 May-2004